

Formative Activity – SWE6302

Based on the "week5-multivariate.ipynb" code base that is downloadable from Week 5 content on Canvas, attempt the following tasks:

1. Data Exploration and Pre-processing [LO3]:
 - i. Load/import the dataset from diamonds.csv, examine its structure, and print the first 20 rows.
 - ii. Visualise data for features 'cut', 'color', 'price', 'carat' and 'clarity' using appropriate plots or graphs.
2. Model Selection and Evaluation [LO3, LO4]:
 - i. Split the dataset into training and testing sets using an appropriate ratio. For example, split 70% data for training and 30% data for testing.
 - ii. Train each model using the training data and evaluate their performance using appropriate evaluation metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared.

Write a report in (+/- 1000 words) explaining what and how you have achieved the above mentioned tasks with appropriate outputs for each individual tasks. Provide your report with the output to the respective tutor of the class to get feedback on your submission on Week 7 – Consolidation week.

Assessment Deliverables:

You are required to produce a report (+/- 1000 words) that discusses all the above factors in the Tasks.

Formatting requirements:

- References list must include a minimum of five to ten sources. As a MINIMUM the Reference List should include two refereed academic journals and three academic books. Harvard referencing format must be used to credit secondary research sources. In-text citations should be included within your discussion (where relevant) using the author-date format and full reference details should be included in your bibliography.
- Diagrams should be captioned and discussed in the body of your report.
- A table of contents should be included.
- Page numbers should be inserted in the centre of the footer.

- The student ID number be placed in the header of each page.

Submission:

Please submit the report before Week 7 – Consolidation week - to the respective tutor of the class via email with the subject line – “SWE6302: Formative Activity Assignment 2 – [STUDENT NAME]”.

Note:

This is a formative (mock) activity that is closely related to Assignment 2 of the module.